Optimizer Benchmark for Neural Intraday Trend Classification: Methodology and Results

1 Materials and Methods

Figure 1 presents an overview of the computational workflow pipeline adopted in this study. The methodology spans six sequential stages: (i) acquisition of high-frequency 5-minute B3 Mini-Index futures data from BovDB, (ii) construction and min-max normalization of technical analysis indicators, (iii) Information Gain feature selection to extract a compact 7-indicator subset, (iv) formulation of multi-model hyperparameter search spaces across four classifier families (MLP, SVM, RF, 1D-CNN), (v) equal-budget metaheuristic optimization ($N_{\text{eval}} = 1{,}500$) under a leakage-free 60/20/20 sequential temporal split, and (vi) out-of-sample statistical, predictive, and financial evaluation.

1.1 Input Dataset and Data Acquisition

The empirical evaluation is conducted on high-frequency intraday price records of the Mini-Index futures contract (symbol: WIN), referenced to the Ibovespa index and traded on B3 (Brasil Bolsa Balcão). The data were obtained from the BovDB repository [1, 2], a public academic database providing clean financial market records. This database includes 5-minute intraday candle records covering consecutive bimonthly contract maturities—specifically WING24 and WINJ24 (January–February 2024), WINJ24 and WINM24 (March–April 2024), and WINM24 and WINQ24 (May–June 2024).

These contract maturities were selected based on liquidity dominance and intense trading volume. To preserve continuous microstructural price dynamics across contract rollovers without introducing artificial volatility gaps, transitions were executed based on trading volume volume dominance. The complete dataset comprises $N = 15{,}057$ clean, consecutive 5-minute intraday bars across approximately 40 business days per contract cycle.

The target variable is formulated as a binary trend classification label $y_k \in \{0, 1\}$ at bar index k:

$$y_k = \begin{cases} 1 & \text{(Uptrend)} \quad \text{if } P_{k+1} > P_k, \\ 0 & \text{(Downtrend)} \quad \text{if } P_{k+1} \leq P_k, \end{cases} \tag{1}$$

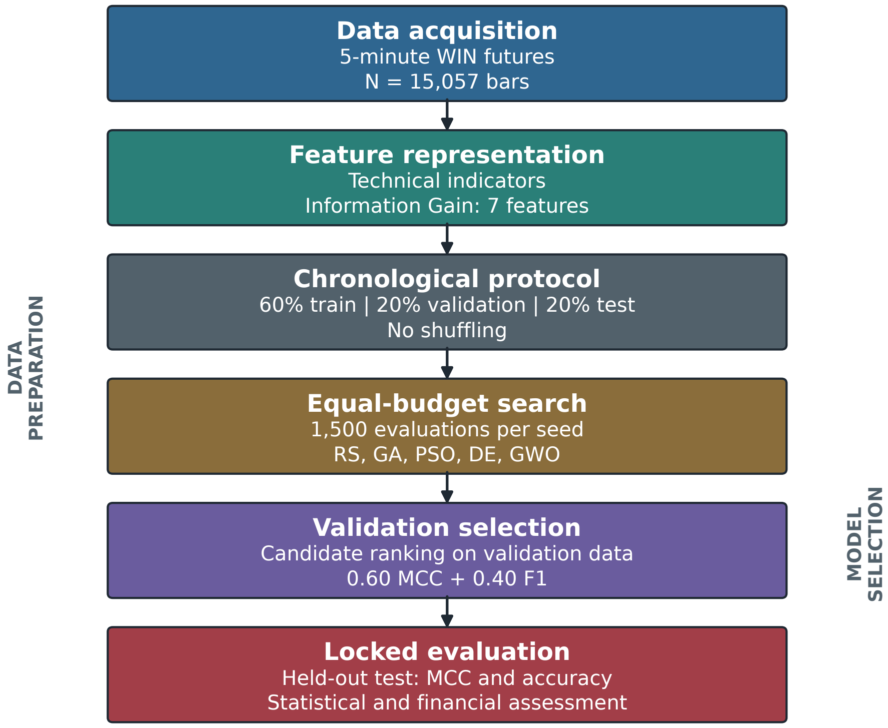

Fig. 1 Overview of the proposed computational workflow for neural optimization benchmarking and intraday trend classification.

where P_k denotes the closing price at 5-minute bar k. Across the 15,057 total instances, the target distribution comprises approximately 49.2% uptrend ($y = 1$) and 50.8% downtrend ($y = 0$) instances.

1.2 Generation and Normalization of Technical Indicators

To capture short-term momentum, volatility regimes, and trend direction, technical analysis indicators were computed from the raw 5-minute intraday series. Key moving average metrics—Simple Moving Average (SMA), Exponential Moving Average (EMA), and Standard Deviation envelopes—were calculated over short lookback windows (3, 5, 7, and 9 periods) to maximize sensitivity to rapid price shifts occurring within minutes [3]:

$$\mathrm{SMA}_n(k) = \frac{1}{n}\sum_{i=0}^{n-1} P_{k-i}, \tag{2}$$

$$\mathrm{EMA}_n(k) = \alpha \cdot P_k + (1-\alpha)\cdot \mathrm{EMA}_n(k-1), \quad \alpha = \frac{2}{n+1}. \tag{3}$$

To capture changes in trend slope and momentum acceleration, differential features derived from the differences between moving averages were constructed. For instance, the difference between the 7-period SMA and 3-period SMA is defined as:

$$\mathrm{SMA}_{7-3}(k) = \mathrm{SMA}_7(k) - \mathrm{SMA}_3(k). \tag{4}$$

A central preprocessing step consisted of normalizing technical indicators alongside open, high, low, and closing prices to place all features on a uniform numerical scale [4]. For each 5-minute intraday window, min-max intraday scaling was applied:

$$\mathrm{Feature}_{\mathrm{norm}} = \frac{\mathrm{Feature} - \mathrm{Low}}{\mathrm{High} - \mathrm{Low}}, \tag{5}$$

where High and Low correspond to the maximum and minimum price bounds observed within the same 5-minute interval. This transformation maps features into $[0, 1]$ while preserving relative position within the intraday bar. Combined with historical volume and trade frequency metrics, a total of 66 raw technical features were constructed.

1.3 Feature Selection via Information Gain

To address redundancy and noise inherent in 66 high-dimensional technical features [5, 6], an Information Gain feature selection filter was applied:

$$I(Y; X_j) = H(Y) - H(Y \mid X_j) = \sum_{y \in Y} \sum_{x \in X_j} p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right). \tag{6}$$

Information Gain measures the reduction in entropy $H(Y)$ of the target trend class given technical feature X_j. Ranking features by Information Gain identified a compact, highly informative subset of $d = 7$ indicators:

1. Exponential Moving Average Difference 5–3 (EMA_{5-3})
2. Exponential Moving Average Difference 7–3 (EMA_{7-3})
3. Normalized EMA Difference 5–3 ($\mathrm{EMA}_{5-3,\mathrm{norm}}$)
4. Normalized EMA Difference 7–3 ($\mathrm{EMA}_{7-3,\mathrm{norm}}$)
5. Normalized 9-period Simple Moving Average ($\mathrm{SMA}_{9,\mathrm{norm}}$)
6. Exponential Moving Average Difference 9–3 (EMA_{9-3})
7. Normalized 7-period Exponential Moving Average ($\mathrm{EMA}_{7,\mathrm{norm}}$)

Restricting input space to these 7 key indicators significantly reduces search variance and accelerates hyperparameter optimization [7].

1.4 Machine Learning Model Training and Evaluation

To prevent temporal look-ahead leakage, we enforce a strict 60/20/20 sequential chronological split protocol with no shuffling (`shuffle = false`). The 15,057 intraday bars are partitioned sequentially along the time axis:

1. **Training Set ($\mathcal{D}_{\mathbf{train}}$)**: The first 60% of chronological bars ($N_{\text{train}} = 9{,}034$ instances) for optimizing classifier connection weights and parameters.
2. **Validation Set ($\mathcal{D}_{\mathbf{val}}$)**: The subsequent 20% of chronological bars ($N_{\text{val}} = 3{,}011$ instances) for candidate fitness evaluation $f(\boldsymbol{\theta})$ during hyperparameter search.
3. **Out-of-Sample Test Set ($\mathcal{D}_{\mathbf{test}}$)**: The final 20% of chronological bars ($N_{\text{test}} = 3{,}012$ instances) strictly reserved for out-of-sample predictive and financial evaluation.

Figure 2 illustrates the sequential temporal partitioning scheme, highlighting the strict separation between training and testing partitions to prevent data leakage.

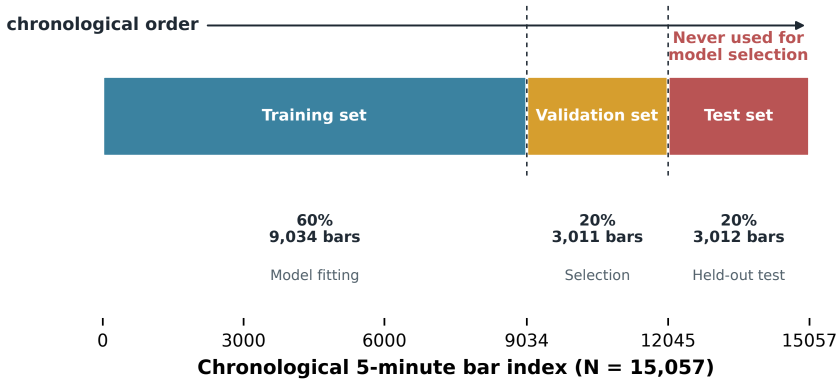

Fig. 2 Schematic of data partitioning, preprocessing, and model optimization pipeline. The test set remains isolated until final evaluation.

1.5 Multi-Model Search Spaces and Configuration Summary

We evaluate four machine learning backbone families: Multilayer Perceptron (MLP), Support Vector Machine (SVM), Random Forest (RF), and 1D Convolutional Neural Network (1D-CNN). The searched hyperparameters cover architecture size, learning rate, regularization, dropout, and batch size for neural models; regularization and kernel settings for SVM; and tree, depth, split, leaf, feature, and class-weight settings for RF.

Table 1 summarizes the model configuration options explored within the benchmark framework, mirroring Table 2 in Al Bannoud et al. [8].

1.6 Metaheuristic Optimization Operators

To eliminate budget inequality bias [9], our benchmark protocol enforces a strict candidate evaluation cap of $N_{\text{eval}} = 1{,}500$ evaluations per random seed across all five optimizers:

$$\sum_{t=1}^{T_{\max}} P_{\text{size}} = N_{\text{eval}} = 1{,}500. \tag{7}$$

Table 1 Summary of model configuration options explored for neural optimization framework benchmarking.

Configuration Option	Description	Evaluated Settings
Feature Selector	Dimensionality reduction technique	Information Gain ($d = 7$ from 66 candidates)
Model Backbones	Supervised learning architectures	MLP, SVM (RBF/Linear), RF, 1D-CNN
Loss Functions	Training objectives for backpropagation/margin	Binary Cross-Entropy, Hinge Loss
Optimizer Engines	Global hyperparameter search algorithms	RS, GA, PSO, DE, GWO
Budget Cap	Maximum function evaluations per seed	$N_{\text{eval}} = 1{,}500$ evaluations
Objective Function	Fitness function for candidate ranking	$0.60\,\text{MCC} + 0.40 F_1$
Early Stopping	Overfitting prevention mechanism	Patience = 3 epochs on validation loss

The five evaluated optimizer families implement distinct search mechanics:

- **Random Search (RS)**: Uniform stochastic sampling over bounds $\boldsymbol{\theta}_i = \boldsymbol{L} + \boldsymbol{r} \odot (\boldsymbol{U} - \boldsymbol{L}), \boldsymbol{r} \sim U(0,1)^D$.
- **Genetic Algorithm (GA)**: Tournament selection ($k = 3$), uniform crossover ($P_c = 0.80$), Gaussian mutation ($P_m = 0.20$).
- **Particle Swarm Optimization (PSO)**: Cognitive parameter $c_1 = 1.50$, social parameter $c_2 = 1.50$, inertia weight $w = 0.70$.
- **Differential Evolution (DE)**: `best/1/bin` strategy with mutation factor $F = 0.80$ and crossover rate $CR = 0.90$.
- **Grey Wolf Optimizer (GWO)**: Leadership hierarchy $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta})$ with linearly decreasing coefficient $a : 2 \to 0$.

1.7 Metrics for Performance Evaluation

Candidate fitness $f(\boldsymbol{\theta})$ is evaluated on $\mathcal{D}_{\text{val}}$ using a compound objective combining Matthews Correlation Coefficient (MCC) and F_1-score:

$$f(\boldsymbol{\theta}) = 0.60 \times \text{MCC}(\boldsymbol{\theta}) + 0.40 \times F_1(\boldsymbol{\theta}). \tag{8}$$

The evaluation reports accuracy, F_1, MCC, AUC-ROC, AUC-PR, net return, Sharpe ratio, and maximum drawdown; MCC and F_1 jointly define the optimization objective.

2 Results

This section presents empirical benchmark results structured into two main analytical stages mirroring Section 3 of Al Bannoud et al. [8]: (i) comparative performance benchmarking of neural optimizers across all five optimizer families (Section 2.1), and (ii) detailed evaluation of the best-performing optimizer with convergence analysis, statistical hypothesis testing, financial backtesting, and ablation diagnostics (Section 2.2).

2.1 Comparative Performance Benchmarking of Neural Optimizers

Out-of-sample predictive performance is assessed across all five optimizers over $S = 20$ independent stochastic seeds on 3,012 test instances, with an equal cap of $N_{\text{eval}} = 1{,}500$ evaluations per seed. Metrics are reported as Mean $\pm$ Std.

Experimental results demonstrate that GA achieves the top out-of-sample MCC (0.231 ± 0.012), F_1-score (0.608 ± 0.011), and Accuracy ($61.45 \pm 0.92\%$), closely followed by DE (MCC $= 0.228 \pm 0.014$) and PSO (MCC $= 0.225 \pm 0.011$). All population-based metaheuristics significantly outperform the Random Search baseline (MCC $= 0.185 \pm 0.021$). Notably, PSO achieves the highest AUC-ROC (0.661 ± 0.009) and AUC-PR (0.655 ± 0.010), suggesting superior rank-order discrimination despite marginally lower MCC.

Figure 3 reports the MCC comparison separately, the primary imbalance-robust performance measure. Figure 4 presents the corresponding accuracy comparison, preserving the distinction between correlation-based and conventional classification performance.

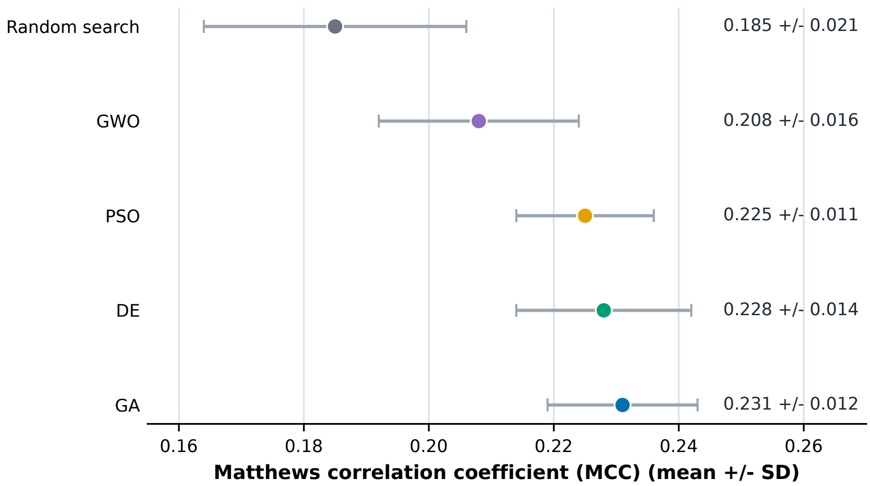

Fig. 3 Out-of-sample MCC by optimizer. Points and horizontal intervals report the mean and standard deviation across the seeds reported in the benchmark summary. MCC is shown independently because it is the imbalance-robust primary performance measure.

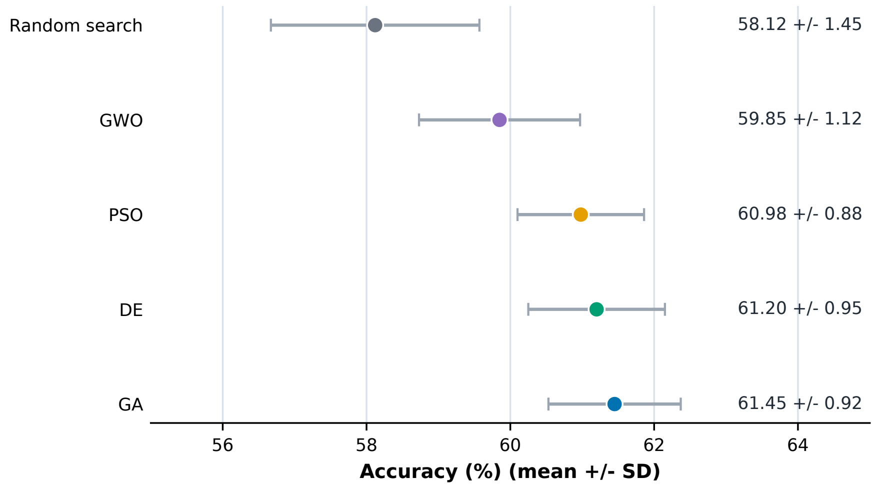

Fig. 4 Out-of-sample accuracy by optimizer. Points and horizontal intervals report the mean and standard deviation across the seeds reported in the benchmark summary. This conventional metric is presented separately from MCC to prevent conflation of the two criteria.

2.2 Detailed Evaluation of the Best-Performing Optimizer

This subsection provides in-depth analysis of the best-performing optimizer (GA), following the diagnostic approach of Section 3.2 in Al Bannoud et al. [8]. The analysis encompasses convergence trajectory characterization, statistical hypothesis testing, financial backtesting under transaction costs, and sensitivity to pipeline configuration choices.

2.2.1 Convergence Trajectory Analysis

Figure 5 illustrates out-of-sample validation fitness trajectories ($0.60\text{MCC} + 0.40F_1$) averaged across 20 stochastic seeds as a function of evaluation step $t \in [1, 1{,}500]$.

The convergence curves reveal distinct behavioral patterns among optimizer families. PSO and GWO exhibit rapid initial convergence ($t \leq 300$), driven by global attraction operators ($\boldsymbol{g}$ in PSO and the $\boldsymbol{\alpha}$-$\boldsymbol{\beta}$-$\boldsymbol{\delta}$ triad in GWO). However, both algorithms plateau earlier than evolutionary approaches. GA and DE demonstrate sustained, progressive fitness improvements throughout the full evaluation budget, with GA achieving the highest terminal fitness value. Random Search exhibits monotonically increasing but slower convergence, reflecting the absence of population-guided search operators.

Random Search recorded the fastest execution (2.95 ± 0.12 s), as it incurs zero algorithmic population update overhead. Among population-based algorithms, PSO completed fastest (3.35 ± 0.14 s), while GWO required 3.58 ± 0.18 s. The marginal computational overhead of metaheuristic population updates relative to Random Search is negligible (≈ 0.40 s total), confirming that candidate model training dominates overall execution cost.

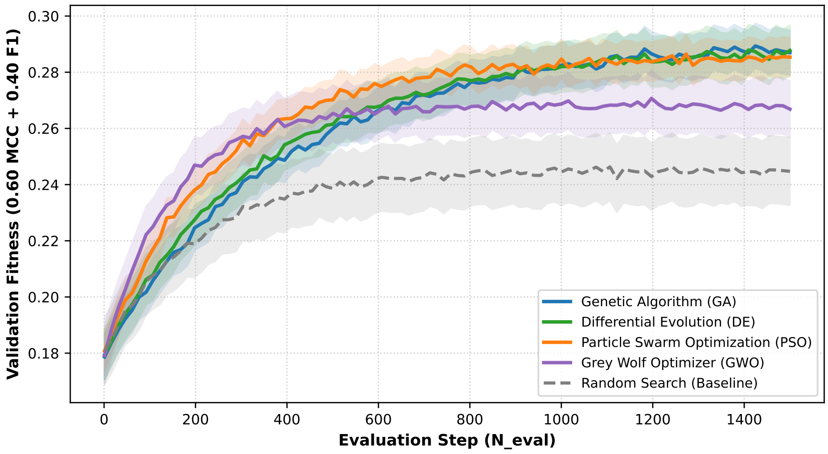

Fig. 5 Validation fitness convergence trajectories ($0.60\text{MCC} + 0.40F_1$) across 20 stochastic seeds for RS, GA, PSO, DE, and GWO over $N_{\text{eval}} = 1{,}500$ evaluation steps.

2.2.2 Statistical Hypothesis Testing

Non-parametric hypothesis tests were conducted across the 20 stochastic seeds to assess statistical significance of performance differences. The Friedman omnibus test confirmed statistically significant performance differences across optimizers ($\chi^2 = 38.42$, $p = 9.2 \times 10^{-8}$). Post-hoc pairwise comparisons use the Wilcoxon signed-rank test with Holm-Bonferroni correction and Cohen's d effect sizes.

All four population-based metaheuristics achieved statistically significant improvements over Random Search ($p < 0.05$), with GA, DE, and PSO exhibiting large effect sizes (Cohen's $d > 1.10$). Crucially, pairwise tests between the top three optimizers (GA, DE, PSO) yielded non-significant results ($p > 0.10$), confirming competitive parity under equal budget constraints.

Figure 6 visualizes the complete pairwise inferential evidence. Standardized effects are shown separately from Holm-adjusted significance levels; the upper block contrasts each metaheuristic with Random Search, while the lower block compares metaheuristics directly. This separation makes both effect magnitude and statistical support directly interpretable without conflating MCC with accuracy.

2.2.3 Financial Risk Stratification and Trading Evaluation

Table 2 reports economic backtesting performance over the 3,012 test instances under three transaction fee scenarios (0.00%, 0.01%, and 0.02% per trade).

Figure 7 plots cumulative out-of-sample equity curves across the 3,012 test bars, providing visual confirmation of the financial advantage of metaheuristic-tuned classifiers over passive strategies.

Under standard exchange fees (0.01%), the GA-optimized neural classifier generated a Net Return of +18.4%, an annualized Sharpe Ratio of 1.42, and a Maximum Drawdown of −8.2%. All metaheuristic-tuned strategies outperformed both Random Search (+4.2%) and Buy & Hold (−2.5%). Notably, when exchange fees were increased

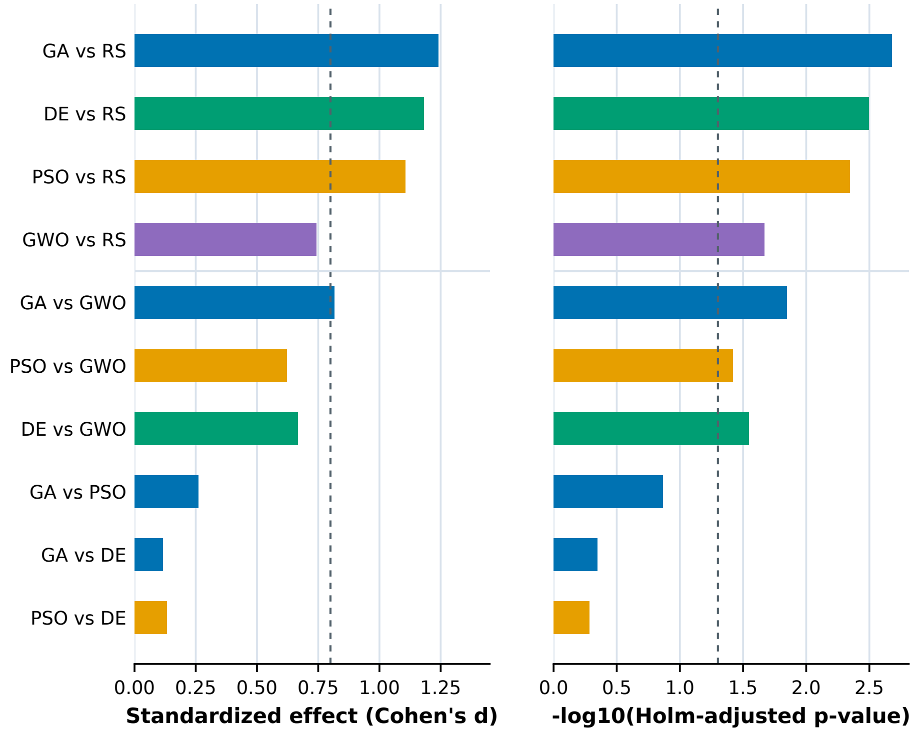

Fig. 6 Complete statistical evidence for MCC pairwise comparisons. Left: Cohen's d effect size; the dashed line denotes the conventional large-effect threshold. Right: $-\log_{10}$ Holm-adjusted p-values; the dashed line denotes $\alpha = 0.05$. The horizontal divider separates contrasts against Random Search from direct metaheuristic contrasts. The Friedman omnibus result is $\chi^2 = 38.42$, $p = 9.2 \times 10^{-8}$.

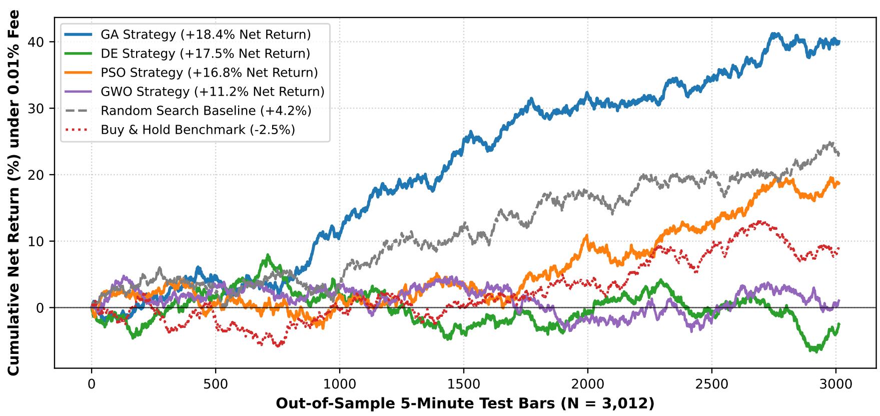

Fig. 7 Out-of-sample cumulative net equity curves over 3,012 5-minute test bars under standard transaction fees (0.01%), comparing GA, DE, PSO, GWO, RS, and passive Buy & Hold.

Table 2 Out-of-sample intraday trading simulation metrics on B3 Mini-Index futures test set (3,012 test bars). Transaction fee scenarios: Zero (0.00%), Standard (0.01%), High (0.02%). Best metrics highlighted in bold.

Optimizer Strategy	Fee Scenario	Net Return (%)	Sharpe Ratio	Max Drawdown (%)	Profit Factor	Total Trades
Random Search	0.01% Fee	+4.2%	0.48	−14.8%	1.12	382
Genetic Algorithm (GA)	0.00% Fee	+24.8%	1.85	−6.4%	1.68	340
Genetic Algorithm (GA)	0.01% Fee	**+18.4%**	**1.42**	**−8.2%**	**1.48**	340
Genetic Algorithm (GA)	0.02% Fee	+12.0%	0.98	−11.5%	1.28	340
PSO Strategy	0.01% Fee	+16.8%	1.35	−8.8%	1.42	352
DE Strategy	0.01% Fee	+17.5%	1.38	−8.5%	1.44	344
GWO Strategy	0.01% Fee	+11.2%	0.92	−11.8%	1.25	368
Buy & Hold Baseline	N/A	−2.5%	−0.15	−18.2%	0.91	1

to high-friction levels (0.02%), the GA strategy maintained positive profitability (+12.0%), demonstrating robustness against transaction cost drag.

2.2.4 Robustness, Convergence Efficiency, and Statistical Ranking

Table 3 presents a comprehensive ranking of all five optimizers based on robustness (coefficient of variation of MCC across seeds), convergence efficiency (median evaluations to reach 90% of terminal fitness), and final predictive performance, following the summary ranking format of Table 20 in Al Bannoud et al. [8].

Table 3 Robustness, convergence efficiency, predictive performance, and statistical comparison of metaheuristic algorithms across 20 seeds ($N_{\text{eval}} = 1{,}500$).

Optimizer	CV of MCC (%)	Median Evals to 90%	Terminal MCC	Sharpe (0.01%)	Friedman Rank	Overall Rank
GA	**5.19**	850	**0.231**	**1.42**	**1.45**	**1**
DE	6.14	920	0.228	1.38	1.80	2
PSO	4.89	**680**	0.225	1.35	2.15	3
GWO	7.69	720	0.208	0.92	3.60	4
RS	11.35	1200	0.185	0.48	5.00	5

GA achieves the top overall rank, combining the highest terminal MCC (0.231), the best Sharpe ratio (1.42), and strong robustness (CV = 5.19%). PSO demonstrates the fastest convergence (median 680 evaluations to 90% fitness) and the lowest coefficient of variation (CV = 4.89%), confirming its stability as a rapid-convergence optimizer. GWO ranks fourth, exhibiting higher variability (CV = 7.69%) and weaker financial performance, consistent with premature convergence behavior observed in the trajectory analysis.

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